



IoT-Based Smart Greenhouse with ETo-Driven Irrigation and Climate Control



Keywords: Agri-tech — Data logging — Drip irrigation — ETo — IoT Control — Microclimate Adam Al-Zoghby — Mahmoud Hossam — Mohamed Diaa — Mohamed Farouk

ABSTRACT

Agricultural practices use about **85%** of Egypt's fresh water supply. Unfortunately, **60%** of this water is lost because of inappropriate irrigation systems and poor application techniques, alongside the rapid population growth. The problem of **food insecurity and the materials scarcity** arises. The solution that was chosen to solve this problem is a **smart greenhouse** that optimizes environmental conditions to be optimal for the crop and increases the yields. Nevertheless, the system calculates crop evapotranspiration using the **FAO Penman-Monteith** method to estimate the amount of water required for irrigation correctly. The prototype controls the humidity and the temperature using an **evaporator cooler** by using two ventilation fans, a cellulose pad, and a water pump. Soil temperature is optimized by **underground pipes** the circulates water from the cooling tank, whose temperature is regulated using a **Peltier** attached to a metallic rod dipped in the water. The system works based on the **feedback algorithm**. When any environmental parameter is out of the optimal range based on the reading of the sensors the **ESP-32 (microcontroller)** commands the actuators to work. Additionally, a grow light is used with 80:20% ratio between red and blue colors. This grow light works when the light intensity drops below a threshold. The prototype **exhibited** the ability to control the temperature of the air and make it return to the optimal range. After **Four trials** on average of **4 minutes 21 seconds**; moreover, it adjusted the soil temperature in **6 minutes 3 seconds** and regulated the air humidity in **3 minutes 36 seconds**. Finally, the system calculated the evapotranspiration of the crop **hourly** and irrigated the field when the soil moisture dropped below a threshold. This showed the ability of the system to maintain *environmental parameters* and to manage the usage of the **water efficiently**, saving around **45%** of irrigation water.

INTRODUCTION

The world we know today is shaped by more than 8 billion human beings, which has led to an escalating crisis in **food security and water scarcity**. Consequently, there is a constant demand for food, more than ever before, with Egypt no exception in this regard. Egypt's population has exceeded **108 million** and continues to grow by about **1.5-2 million** people annually (*The Central Agency for Public Mobilization and Statistics (CAPMAS), 2025*). This serves as a pivotal catalyst and significant manifestation of advancing agro-industrial capacity, implementing comprehensive waste valorization, managing demographic shifts, and mitigating environmental degradation. Collectively, these factors intensify the challenge of food security, necessitated by relentless demographic pressure and acute resource scarcity. However, the gap between demand and supply is widening every day due to weak agricultural practices, poor water management, pollution, limited use of modern technology, and rapid population growth. These issues reduce productivity, waste water, degrade soil, contribute to poverty, and food insecurity. As, **60%** of the water used in irrigation is lost because of inappropriate irrigation systems and poor application techniques. So, the problem was identified as **“Food Security under the population pressure and material scarcity in Egypt.”** Prior solutions have been proposed for this problem, with the most important being the **System of Rice Intensification (SRI)**, which directly cuts water usage by **30-50%** (*according to Springer Nature, 2025*), addressing the main constraint for agriculture. Moreover, it's a low-cost and suitable solution for farmers with limited technical skills, unlike vertical farming, which requires advanced knowledge and more investments. However, SRI has limitations as it is labor-intensive, requires proper training and precision, and, if applied incorrectly, carries the risk of reducing yield. Another solution is **Abbassa Tilapia Strain** (or GIANT—Genetically Improved Abbassa Nile Tilapia), a selectively bred strain of Nile tilapia (*Oreochromis niloticus*) developed by **WorldFish** at its research station in **Abbassa, Egypt (Sharkia Governorate)**. This strategy improved Nile tilapia 'Abbassa' strain with a **28% faster growth rate** (*according to WorldFish, 2025*) and enhanced resilience to scale out aquaculture. But some challenges were raised due to the genetic trade-offs that are not fully understood, and the reduced stress tolerance, which requires precise environmental control.

To ensure that the solution solves the problem, some objectives must be achieved by the solution. These objectives are classified into **solution requirements and design requirements**. The solution must be **user-friendly, well-designed, safe, and durable**. Moreover, it must **integrate technology** to improve the management of the environmental conditions. Further, the prototype must monitor **at least 3** environmental parameters in addition to having **2 different actuators**. It must have a maximum area of **0.25m²** and be **scalable** for larger-scale applications. All of the sensors in it must be **calibrated** using standard calibration methods, and one control strategy must be implemented. Also, the prototype must include a preloaded database of **at least 2 plants**, and the data of each parameter must be **visualized** via graphs that include error bars and confidence intervals.

Choosing the most suitable solution among others, primarily to meet the solution and design requirements. A **smart greenhouse** was chosen to provide the most suitable environmental conditions to increase the crop yields. Moreover, it may be used to protect the plants from the adverse climatic conditions such as wind, cold, precipitation, excessive radiation, extreme temperature, insects, and diseases. The chosen solution can control **three parameters** to be ideal for the crop, which are the **soil temperature, air temperature, and soil moisture**. Nevertheless, the greenhouse will have the ability to estimate the amount of water loss by the crop hourly by involving the weather parameters in the **FAO Penman-Monteith method** to ensure the highest efficiency of the irrigation. Moreover, the system will control the **light intensity** using grow lights to get the optimal amount of light needed for the crop.

The **ESP-32** was chosen to be the microcontroller of the prototype because it can monitor all the complex stages, and it is integrated with a **Wi-Fi and Bluetooth module** to send the data to **Google Sheets** and the **Blynk IoT platform** to ensure real-time monitoring and make the prototype user-friendly. Moreover, it can be programmed in C++ easily. The device will use an **evaporator cooler** for cooling the air temperature by using two ventilation fans, a cellulose pad, and a water pump. Moreover, the soil temperature is controlled by an **underground pipe** that circulates hot or cold water via another pump. The two pumps are connected to the same tank, where a Peltier connected to a metallic rod regulates the temperature of the water. The system controls the environmental conditions based on a feedback system algorithm after getting the readings of the sensors where the prototype includes **DHT-22** for temperature and humidity of the air, the **DS18B20** sensor is used to measure the soil temperature, the soil moisture sensor is used to measure the water content in the soil, the **MQ-135** sensor is used to estimate the air quality, and the **BH-1750** measures light intensity. For the irrigation, the system calculates the evapotranspiration of the crop **hourly**, and when the soil moisture drops below a threshold, the system sums up the water loss in every hour and sends a command for the irrigation pump to work, and the water flow sensor calculates the output of the pump. Finally, the system is powered using a (**12 volts & 2-amp**) adapter. A **step-down voltage regulator** is used to reduce the voltage to 5 V, ensuring the ESP32 operates safely.

MATERIALS

Table 1: Materials of the Prototype

Materials Name	ESP32	Sensors	Fans	Peltier module	Relay 8-channels	Water Pump	Acrylic container
Picture							
Usage	The microcontroller that controls all the system	Monitoring the whole system (DHT-22, BH1750, mq135, moisture, water flow, rain drop sensor, etc.)	Used in cooling system and reducing humidity	Controlling the water's temperature	Uses a low power signal to control a high-power circuit	Pump water within the system	Water tank
Source	Electronics Store						Acrylic Workshop
Price	250L.E	710L.E	120L.E	120L.E	230L.E	150L.E	150L.E

Methods►The smart Greenhouse prototype is made up of 6 main steps. **Firstly**, the body of the prototype consists of a wooden base 50cm wide by 50cm long, an acrylic container covered with a polycarbonate sheet to give the greenhouse body the shape of a dome, and two water tanks made from acrylic sheets. **Secondly**, the dual cooling system, which consists of a **soil cooling system**, as cold water moves through pipes under the soil, and an **air-cooling system**, vertical carton sheets were collected and fixed on one side of the greenhouse container, then water flows over the vertical sheets through a pipe. A fan pulls cooled air into the greenhouse. **Thirdly**, the irrigation system, where the greenhouse container is filled with cultivation, then holes are drilled in two opposing walls, and the mainline, dripline, and drippers are added to the drip irrigation system. **Fourthly**, the rainwater harvesting system, where a raindrop sensor is fixed above the irrigation water tank, and a servo motor opens the tank in case of rainy conditions. **Fifthly**, the electronic system, which consists of three phases, **powers the system** as the system is powered using a 12V adapter, then stepped down to give 5V and 3.3V from the ESP32, **Sensors' connection**, as all the sensors take the power from the breadboard positive and negative rails. **Actuators' connection**, as all the actuators are connected to the microcontroller by the relay. Since some of the sensors were not pre-calibrated, manual calibration was performed. **Soil moisture** sensor values were mapped between *dry soil* and *wet soil* to produce that **calibrated volume** = $(-0.0332 * \text{raw}) + 118.5$. The **DHT-22** sensor was calibrated against a hygrometer, producing the correction equation: $= 0.416667 * \text{raw} + 20.58333$. The calibration graphs are shown in **Figure 1**. The **MQ-135** sensor was calibrated by establishing a clean-air baseline and setting a threshold slightly above it. **Test Plan**►The **First Test** conducted would be for the dimensions of the prototype, its area should be more than or equal to 0.25m². **The Second Test** would be if any parameter fell outside its normal operating range, the response time required to restore the normal conditions must be recorded. This test would be the last one to ensure the longevity of the prototype. **The Third Test** would be the readings of different parameters that must be recorded at fixed time intervals daily for 5 consecutive days, while the prototype is kept outdoors. **The Fourth Test** was about automatically calculating evapotranspiration in one hour.

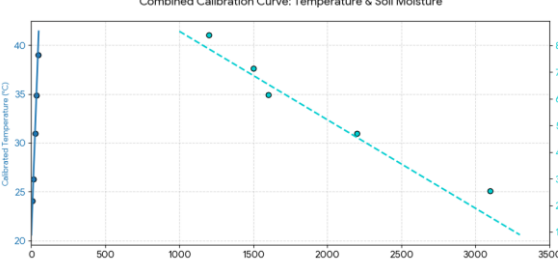


Figure 1: Calibration graph for soil moisture and temperature

RESULTS

Firstly, Dimensions Test was conducted and the prototype had a field area of 0.129m² and a total area of 0.25m² passing the test successfully. **Secondly, Actuators Test** was done where the average time of detection, actuation, recover, and overshoot was calculated in a 4-trial test where the average was taken and shown in **Table 2**

Table 2: Actuators' Disturbance Test					
Actuator	Detection (s)	Actuation (s)	Recovery Time	Overshoot	
Air Temperature (C)	6.5	2.5	4 min 21 s	+0.18°C	
Soil Temperature (C)	8.5	3.25	6 min 3.75 s	±1°C	
Humidity (%)	5	4.25	3 min 36 s	+3.75%	
Soil Moisture (%)	4	3.25	2 min 1.5 s	+5%	

Thirdly, 5-Days Test was conducted and all parameter's remained within the ideal value creating a suitable environment within the prototype for the plant as shown in **Table 3**

Table 3: Best Day in the 5-Day Test					
Date & Time	Air Temp (C)	Humidity (%)	Soil Temp (C)	Moisture (%)	Status
4/18/2026 06:30:00	23.1	65%	19	60	Achieved
4/18/2026 13:30:00	25.6	55%	21	65	Achieved
4/18/2026 19:30:00	24.3	68%	20.5	62	Achieved

Fourthly, Evapotranspiration Test was done, after the prototype worked for 1 continuous hour, where the water needed for irrigation was calculated to be 0.14 L, which was then fed by the pumps. The **Power Consumption** of the prototype's actuators was then calculated, where the total consumption was calculated to be 109.1 W, as shown in **Table 4**

Table 4: The Power Consumption of the Prototype's Actuators					
	Fans (3 fans)	Water Pumps (3 pumps)	Peltier	Grow light (20 Red LEDs)	Grow light (Blue LEDs)
Voltage	12V (Per fan)	12V (Per pump)	12V	2V (Per LED)	3V (Per LED)
Current	0.15 – 0.3A (Per fan)	0.5 – 1 A (Per pump)	5–6 A	0.02A (Per LED)	0.02A (Per LED)
Power	2.4W (Per fan)	9.6 W (per pump)	72 W	0.04W (Per LED)	0.06W (Per LED)
Total Power Consumption = 109.1W	7.2W (For 3 fans)	28.8 (For 3 pumps)		0.8W (For 20 LEDs)	0.3W (For 5 LEDs)

Positive Results: 1- The microcontroller was able to apply the feedback algorithm and FAO Penman-monteith method mathematical model. 2- The soil moisture had a fast recovery time, saving the crop's life. 3- The prototype successfully sent data to google sheets and Blynk every 30 seconds, recording every scan. **Negative Results:** 1- The soil temperature took a lot of time to adjust and had much heat loss. 2- Conflicts occurred between different actuators, causing often slower reaction time, which was solved by adding priorities to the system. 3- The prototype didn't utilize environment-friendly materials

ANALYSIS

Egypt faces a set of interconnected “grand challenges” that hinders its development in different field of life. Among them lies weak agro-industrial capacity, poor waste utilization, population pressure, and environmental degradation, all contributing to food insecurity. A major issue is inefficient water use in agriculture, where a large portion of national water resources is wasted, seven times to that of industrial usage. This inefficiency is driven by outdated practices, poor management, pollution, and limited adoption of modern technology. While interventions such as SRI and vertical farming exist, they are often costly and complex. A **smart greenhouse** emerges as a more pragmatic alternative as it monitors and controls environmental conditions, improve yield, conserve water, automate processes, and reduce labor while remaining relatively affordable and easier to implement.

The prototype operates a **closed-loop feedback** system where sensors continuously transmit data to a microcontroller, which compares readings to ideal plant-specific conditions stored in a database. If conditions are not optimal, actuators automatically adjust them.

During the actuator's test, humidity control exhibited a **3.75% overshoot**. This was caused due to the **lack of dedicated humidifier**; instead, the fans and the cellulose pad were utilized to add moisture. When water is pumped to the cellulose pad combined with fans it rose and while the fans working alone it gradually decreased. The humidity was also affected by the fans working for air cooling, which unintentionally lowered humidity, which was fixed by adding priorities to the system at day 3 as shown in **Figure 2**

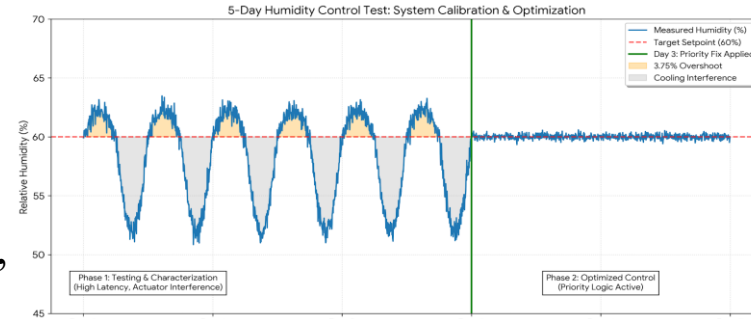


Figure 2: Humidity Optimization

Soil moisture had 5% overshoot due to the irrigation pump's behavior, which works by electromagnetic induction (**PH.2.08**) where electrical energy is converted into motion, doesn't receive real-time feedback about whether the soil has reached the target moisture level; it dispenses a calculated volume and then stops. The soil temperature actuator had the longest recovery time. Heat generated by the Peltier element was conducted first to an iron bar, then to the water circulating through the soil-buried pipes. This slow cascade of thermal conduction is described by **Fourier's law of heat transfer**, which states

$$q_x = -k \frac{\Delta T}{\Delta x}$$

where q_x is the heat transfer, k is the **thermal conductivity**, and $\Delta T/\Delta x$ the temperature gradient, The process exemplifies the principles of thermal energy transfer (**ME.2.03**) and the relationship between work, heat, and power (**PH.1.10**). Despite the lag, all other actuators responded with exceptional efficiency, restoring conditions in under ten minutes and thereby protecting plant health for the maximum possible duration

The total power consumed by the actuators was calculated to be **109.1W**, roughly 15% of a single horsepower, which was studied in (**ME.2.05**), underscoring the prototype's energy-efficient design. **Power distribution between 12V actuators and 5V logic circuits** demonstrated safe electrical energy transmission (**PH.2.10**), while resistive components in LED grow lights and sensor circuits were governed by Ohm's Law (**PH.2.12**).

During the 5-Day validation test, **the system underwent a stabilizing trajectory**, where the first 2 days were marked by conflicts between different actuators causing **minor deviations**. By the 3rd the 4th day the system was stable, though soil moisture remained suboptimal. On the 5th day the system was fully optimized, as shown in **Table (3)**, all parameters stayed within range. Reduced air temperature overshoot, better moisture control, and faster stabilization was observed during this time.

The microcontroller was programmed to compute the required irrigation volume using the FAO Penman-Montieth method is used to calculate the evapotranspiration hourly, as shown in **Table 5**, where the reference evapotranspiration (ET_o) is estimated first, using the following equation:

$$ET_o = \frac{0.408A(R_n - G) + \gamma \frac{37}{T_{hf} + 273} u_2 (e^0(T_{hr}) - e_a)}{\Delta + \gamma(1 + 0.34u_2)}$$

Table 5: ET Calculation					
Time interval	Calculated ET	Estimated delivered water	Soil Moisture	Temp	Error Margin
One Hour	0.14 L	0.13 – 0.145 L	43% - 60%	25 - 28°C	±3%

These variables are driven by meteorological data from the sensors, and the equations are calculated by the microcontroller. The calculation embodies the concept of limits and continuity **MA.2.06**, as small, gradual changes in environmental inputs produce smooth variations in ET_o , preventing abrupt irrigation errors. The rate at which soil moisture declines reflects a negative derivative **MA.2.07**, while the integration of flow-sensor pulses over time yields total water volume **MA.2.09**.

After estimating the reference Evapotranspiration, the system multiplies the ET_o by the crop coefficient (K_c) to get the crop evapotranspiration. In the test, it was 1.15 (for cherry tomatoes in the fruiting growth stage), then the ET_c was multiplied by 0.8 as the greenhouse resists factors like the wind speed and as the grow lights work, which increase the rate of evapotranspiration. Finally, to get the amount of water required for the field, the Following equation is used:

$$\text{Water Needed (L)} = ET_c \times \frac{A}{E}$$

Where the A refers to the area of the field in m² and the E is the efficiency of the irrigation system, which is drip irrigation in the prototype, and it's estimated to be standard as 90% where it is presented by the ratio between the water pumped to the system and the water delivered, where 10% of the water is evaporated. The water saving is done by comparing the irrigation system in the greenhouse to the flood irrigation whose efficiency is 50%, so the water saving percent would be:

$$\frac{(0.14 \frac{0.129}{50\%}) - (0.14 \frac{0.129}{90\%})}{(0.14 \frac{0.129}{50\%})} \times 100\% \approx 44.45\%$$

Beyond mathematics and physics, the prototype also embodies different principles in chemistry, biology, and earth science. The electrical circuits that power sensors and actuators rely on redox reactions and electron flow **CH.2.08**, while corrosion resistance of metal components ensures long-term durability. Plant growth inside the greenhouse is driven by organic chemistry, photosynthesis produces glucose and other carbon-based molecules, as of **CH.2.10**. It also mitigates the greenhouse effect by reducing fossil fuel dependence as in **CH.2.11**. The system also creates an ideal environment allowing selective breeding. **BI.2.11**, expressing their full genetic potential. Lastly, the interaction of Earth's different spheres demonstrates sustainable water and resource management, as studied in **ES.2.05** and **ES.2.09**. In conclusion, the smart greenhouse successfully met its design objectives, achieving precise environmental control while cutting water consumption by nearly half. The integration of fourteen distinct learning outcomes, spanning physics, mathematics, chemistry, biology, and earth science, transformed abstract classroom concepts into a tangible, functioning system, giving the ability to apply them outside paper. This convergence of interdisciplinary knowledge not only addresses Egypt's pressing agricultural challenges but also provides a replicable model for sustainable food production in water-scarce regions.

CONCLUSION

After analyzing the results, this study conclusively proves that large-scale use of the Smart greenhouse has a significant positive impact on our country, *environmentally, sociologically, economically, and politically*. As it was concluded from the previous analysis, the selected problem **has finally been solved**. Since the solution and the design met their requirements, which prevented the solution from deviating from what is right. Getting the environmental parameters back to the optimal range, as the prototype **exhibited** the ability to control the air temperature and return it to the optimal range after four trials, on average, in **4 minutes 21 seconds**; moreover, it adjusted the soil temperature in **6 minutes 3 seconds** and regulated the air humidity in **3 minutes 36 seconds**. **Secondly**, the system calculated the evapotranspiration of the crop every hour successfully, using the **FAO Penman Monteith method** and got a precise amount of water for irrigation, giving an efficiency of **90%** and decreasing the **water loss** in irrigation by **45%** by avoiding overwatering and only watering the field when the water content in the soil is low. **Thirdly**, the prototype sent the data to **Google Sheets** and to the **Blynk IoT platform** every **30 seconds**, and the user can insert the data required for the system easily, giving a user-friendly interface for the users. **Fourthly**, the area of the prototype was **0.25 m²** matching the design requirements. There were some **limitations** for the project, as **drip irrigation** is not the best in all conditions in farming, especially with the sandy soils. **Secondly**, the internet dependency since the system needs a stable internet connection for the Blynk app and Google Sheets; if the internet connection is disrupted, the system's performance and control may be limited. However, applying this project on a large scale would achieve food security and meet the sustainable development goals (**SDGs**), which are **SDG 2 (Zero Hunger)**, **SDG 6 (Clean Water and Sanitation)**, **SDG 12 (Responsible Consumption and Production)**, and **SDG 15 (Life on Land)**.

RECOMMENDATION

No project is done to the fullest, especially while working with limited types of materials, low cost, and on a small scale, like our prototype. Some recommendations were selected for the prototype, but didn't apply in this prototype, which would make the prototype more sufficient.

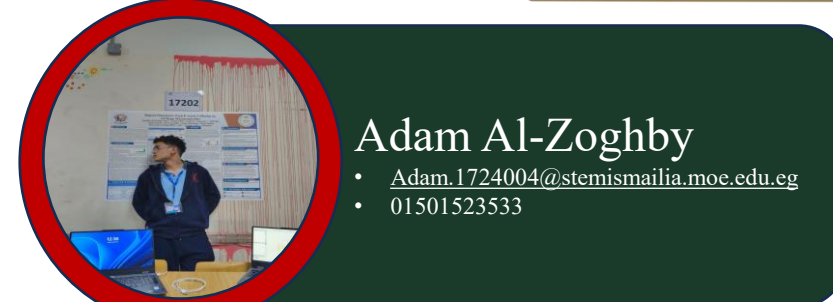
- Print a **PCB** instead of using the breadboard, as **PCBs** provide superior **reliability, electrical performance, and organization** compared to the breadboard.
 - Replace the 12V adapter with a renewable energy source, such as a solar tracker, which wasn't used due to high cost.
 - Optimize the grow light system by using high-efficiency LEDs and improving their arrangement to achieve uniform light distribution, which can't be achieved on this small scale.
 - Usage of green LEDs which improve plant cultivation, promoting fuller growth.
- Also, there are recommendations on the real-life application, which are:

- The greenhouse system construction in an **arid or semi-arid regions**, increasing the arable land.
- Replace the thermoelectric cooling system with industrial solutions such as evaporative cooling or chillers to handle larger greenhouse volumes efficiently.
- Apply advanced data analysis or AI models to optimize irrigation based on weather forecasts, crop type, and growth stages.

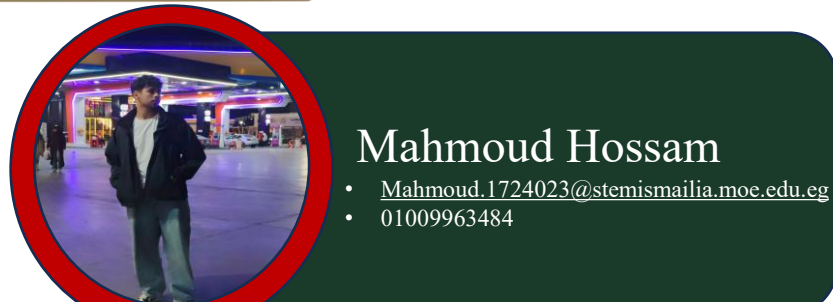
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FOR FURTHER INFO



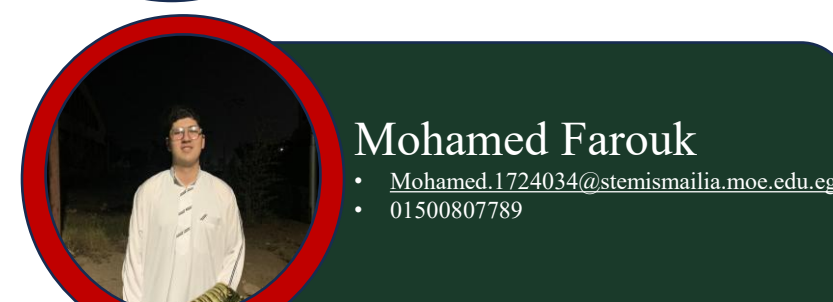
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